🌺 Yagé Botanicals Cloud Migration Proposal

Yagé Botanicals

Network Modernization Proposal

Prepared by 0x2A Security

Analyst: Brandon Cavalier

Date: 8/10/2025

👑 Step 1: Identify assets

Step 1: Identify assets.

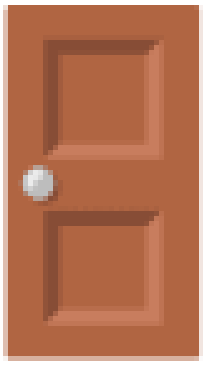
Establishment of a Configuration Management Database (CMDB) in Alignment with CIS Controls

The initial step involves the systematic identification and comprehensive cataloging of the organization's most critical systems, following the guidance of CIS (Center for Internet Security) controls. This process requires an organized and thorough inventory of all network assets, including servers, workstations, and network-connected devices. Asset documentation should encompass detailed data on hardware specifications, operating systems, IP addresses, and the software applications installed on each device. All information must be recorded within an IT Operations (ITop) or equivalent asset management system to enable routine updates, maintenance, and audit compliance.

Assets should be classified according to parameters such as their organizational value, operational criticality, and data sensitivity. This classification supports the implementation of prioritized security measures and risk management strategies. The inventory should also include a comprehensive catalog of all software applications and their versions, with particular emphasis on identifying outdated or unsupported software requiring patching or upgrades.

Following asset documentation, a detailed network diagram must be developed. This visual representation should depict all network-connected devices, illustrating their physical locations, interconnections, and network segmentation—including subnetworks and VLANs where applicable. Precise depiction of each device's IP address, network segment, and connectivity details will provide a comprehensive overview of the network topology, facilitating effective security management and incident response planning.

🚪 Step 2: Implement basic security measures



Step 2: Implement basic security measures

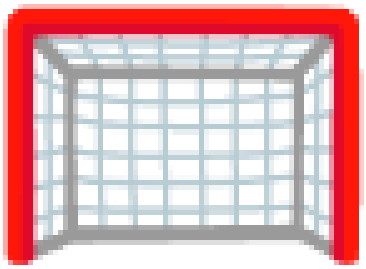
To enhance network security, implement network segmentation by utilizing VLANs to separate critical systems from other network segments. This segmentation reduces the risk of lateral movement by isolating sensitive assets from less secure segments.

For Bring Your Own Device (BYOD) users, employ a Software as a Service (SaaS) mobile device management (MDM) solution. A cloud-based service such as Microsoft Intune can be used to enforce security policies automatically, enable remote wipe capabilities for lost or stolen devices, control application installations and updates, and monitor compliance with company policies.

Adopt the principle of least privilege by ensuring that access to critical systems is restricted to authorized personnel only. Additionally, implement Multi-Factor Authentication (MFA) on all devices connected to the network to provide an extra layer of security.

Regularly update firewalls, apply necessary patches, and ensure software is kept current to address vulnerabilities and maintain system integrity.

🥅 Step 3: Strategize the migration



Step 3: Strategize the migration.

Establish a Public cloud architecture using Google Cloud Platform (GCP). Addressing the challenge of outdated technology discouraging potential buyers requires a strategic approach focused on modernization without incurring excessive capital expenditure. By leveraging Google Cloud Platform (GCP), Yagé Botanicals can effectively transform its technological landscape, enhancing its appeal to prospective buyers. Shifting from on-premises infrastructure to GCP's infrastructure-as-a-service (IaaS) and platform-as-a-service (PaaS) models is key. This transition moves expenses from a capital expenditure model to an operational expenditure model, allowing Yagé Botanicals to pay only for the resources consumed, thus reducing initial investment. GCP's scalability and flexibility enable the business to dynamically adjust resources based on demand, ensuring efficient resource allocation and preventing wasteful expenditure on unused capacity. The pay-as-you-go pricing model inherent in GCP's services further contributes to cost-effectiveness by eliminating substantial upfront licensing fees and enabling precise cost optimization through consumption-based billing.

GCP provides access to a comprehensive suite of modern technologies, including artificial intelligence, machine learning, advanced data analytics, and serverless computing. By integrating these cutting-edge solutions, Yagé Botanicals can significantly modernize operations without the burden of investing in and maintaining expensive on-premises systems. This technological upgrade improves the agility and speed of application and service deployments, facilitating quicker responses to market dynamics and customer demands. Additionally, GCP offers robust security features and a wide array of compliance certifications, strengthening the security posture and regulatory compliance of the organization. By leveraging services such as Compute Engine for migrating workloads to virtual machines, Cloud SQL for replacing legacy databases with managed services, App Engine or Cloud Functions for modernizing applications with

microservices architecture, and BigQuery for implementing cloud-based data warehousing,Yagé Botanicals can enhance its value proposition without substantial financial risk. The strategic adoption of GCP not only modernizes the technology infrastructure but also increases operational efficiency and significantly improves the company's attractiveness to potential buyers.

🕵️ Step 4: Establish ongoing security monitoring

Step 4: Establish ongoing security monitoring.

Establishment of a Security Orchestration, Automation, and Response (SOAR)-Enhanced Hybrid Security and Network Operations Center (SNOC)

Given the increasing complexity of modern IT environments, characterized by interconnected systems where network issues can rapidly escalate into security threats and the blurring lines between infrastructure and security in cloud and virtualized environments, the establishment of a hybrid Security and Network Operations Center (SNOC) augmented by Security Orchestration, Automation, and Response (SOAR) capabilities is recommended. This integrated approach addresses the critical need for faster incident response while optimizing resource utilization and minimizing operational costs.

The integration of network and security monitoring into a unified SNOC provides a holistic view of infrastructure health and security threats, enabling rapid identification and resolution of incidents. This streamlined approach eliminates communication delays between separate teams, facilitating a more coordinated and effective response. By consolidating expertise and resources, the SNOC optimizes resource allocation and reduces the need for specialized handoffs between network and security teams, leading to improved resource utilization and operational efficiency.

The adoption of a hybrid SNOC architecture offers significant cost advantages, including the consolidation of personnel into a single, cross-functional team and the sharing of tools and infrastructure. This reduces the overhead associated with maintaining separate network and security operations centers, leading to optimized staffing levels and more efficient resource allocation.

To effectively implement the SOAR-enhanced hybrid SNOC, the integration of diverse technology platforms is essential. This includes network monitoring platforms such as SolarWinds and Nagios, security tools such as Security Information and Event Management (SIEM) platforms and Intrusion Detection/Prevention Systems (IDS/IPS), and cloud platforms such as Google Cloud Platform (GCP). Scripting and automation tools further enhance the capabilities of the SNOC by automating routine tasks and enabling rapid incident response.

💼 Step 5: Stabilize ongoing business operations

Step 5: Stabilize ongoing business operations.

To secure the continuity of ongoing business operations, the strategic implementation of three pivotal policies is essential. These initiatives address critical areas of operational risk, encompassing incident response protocols, security monitoring practices, and infrastructure maintenance procedures. First, a comprehensively updated Incident Response Plan, or IRP, is paramount. The updated IRP should offer clear pathways for incident escalation, clearly defined roles and responsibilities, comprehensive communication protocols, and specific action plans tailored to address a range of potential incident scenarios. To ensure its effectiveness, the IRP should undergo routine testing and be updated to reflect the evolving threat landscape and organizational changes. Secondly, a policy of continuous auditing of passwords, logins, and network traffic should be established. This practice is vital for the early detection and prevention of unauthorized access attempts and potential malicious activity. The implementation of automated monitoring tools that track user authentication attempts, changes in passwords, and variations in network traffic patterns is necessary for the success of this initiative. Any anomalous activity detected should trigger immediate alerts for further investigation. Finally, the establishment of a defined maintenance window for the consistent performance of software patching, CMDB updates, and regular firewall calibrations is recommended. This policy requires a predetermined timeframe during which these necessary activities can occur. In addition, rigorous testing of updates and configurations prior to implementation is crucial to mitigate the risk of unintended consequences. By implementing these three policies in a concerted and coordinated manner, organizations can substantially enhance the stability and security of their ongoing business operations, effectively reducing the likelihood of disruptive incidents and ensuring the continuation of essential business activities.